Prompt Ladder Assignment

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Track: Machine Learning Engineering

Project: IntelliPlant

This assignment demonstrates how a weak prompt evolves into a reusable prompt by adding exactly one improvement layer at a time and observing the effect on the model's output.

Baseline

Layer Added	None
Prompt	Build a machine learning pipeline.
Output Improvement	Claude asked clarifying questions because the request was too vague.
What Still Failed	Prompt lacked enough context to generate a pipeline.
Next Change	Add a clear goal.

Version 1

Layer Added	Clear Goal
Prompt	Build a machine learning pipeline for predicting plant health and disease risk.
Output Improvement	Generated a task-specific ML pipeline instead of asking only questions.
What Still Failed	Still assumed project details.
Next Change	Add project context.

Version 2

Layer Added	Context
Prompt	Previous prompt + 'This is for my IntelliPlant project.'
Output Improvement	Recommendations aligned with the IntelliPlant use case.
What Still Failed	Output remained paragraph-heavy.
Next Change	Specify an output format.

Version 3

Layer Added	Output Format

Prompt	Previous prompt + 'Present the answer as numbered implementation steps.'
Output Improvement	The response became a structured checklist.
What Still Failed	Reasons behind the steps were limited.
Next Change	Ask for explanations.

Version 4

Layer Added	Quality Criteria
Prompt	Previous prompt + 'Explain why each step is important.'
Output Improvement	Each step included practical reasoning and design justification.
What Still Failed	Response became longer than necessary.
Next Change	Add constraints.

Version 5

Layer Added	Constraints
Prompt	Previous prompt + 'Keep the answer under 500 words.'
Output Improvement	Concise and easy to follow.
What Still Failed	Some useful technical detail was lost because of the word limit.
Next Change	Final reusable prompt.

Final Reusable Prompt

Build a machine learning pipeline for predicting plant health and disease risk for my IntelliPlant project. Present the answer as numbered implementation steps. Explain why each step is important, use beginner-friendly language, recommend the most practical MVP approach, and keep the response concise enough to use as a development checklist.

Reflection

The largest improvement came from adding a clear goal and project context. Formatting made the response easier to use, while the word limit improved readability but removed some technical detail. This exercise showed that improving one prompt layer at a time makes it easy to identify which changes genuinely improve the output.